



# AI-Driven Attendance and Workflow Management System for Academic and Professional Environments

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## 1. Executive Summary

### 1.1. System Overview

The Agent-Based Intelligent Attendance and Workflow Tracking System represents a sophisticated multi-agent platform designed to transform raw badge-based attendance data into actionable insights for institutional decision-making. Built on the Smol Agents framework, it integrates data preprocessing, anomaly detection, Louvain clustering for cohort analysis, and a React-based decision-support dashboard, enabling seamless deployment across universities, corporate training programs, and professional development environments. This prototype delivers real-time analytics through a FastAPI backend, with specialized agents handling validation, grouping, and insight generation while maintaining persistent memory for contextual reasoning.

### 1.2. Problem Statement

Conventional badge-based attendance systems generate noisy, incomplete logs plagued by duplicates, hardware anomalies, and proxy entries, rendering them inadequate for proactive intervention in dynamic institutional settings [1], [2]. Administrators and supervisors lack interpretable insights into individual behaviors, cohort trends, or at-risk patterns, relying on manual analysis that hinders timely decision-making and strategic resource allocation. This limitation undermines attendance data's potential as a proxy for engagement, performance, and operational efficiency across educational and professional contexts.

### 1.3. Proposed Solution

The system employs an orchestrator agent that routes queries through specialized sub-agents—data pipeline, validation, grouping, and insight generation—to deliver explainable, data-grounded responses without exposing internal logic. It preprocesses badge logs against iCalendar schedules, applies rule-based validation for anomalies, identifies attendance-based clusters via the Louvain algorithm, and surfaces insights through an interactive dashboard featuring engagement matrices, session details, and alert visualizations. This architecture ensures scalability, with background scheduling for autonomous updates and hot-reload configuration, positioning it as a production-ready product for institutional integration [3].

### 1.4. Key Benefits

The proposed system offers institutions a more intelligent and operationally effective approach to attendance management by transforming raw badge data into actionable insights. It reduces administrative burden through automated anomaly detection, cohort discovery, and structured validation, while improving the reliability and interpretability of attendance records. By identifying recurring absences, late arrivals, and unusual behavioral patterns at an early stage, the system supports timely intervention and more effective resource allocation.

Its agent-based architecture further enhances decision-making by providing explainable, query-driven analytics through a unified dashboard. This enables staff to explore attendance behavior at both individual and group levels without relying on manual analysis or static reports. As a result, attendance data becomes a strategic asset for monitoring engagement, supporting performance, and strengthening institutional oversight.

## 2. Introduction

### 2.1. Background and Context

Attendance monitoring has evolved from manual roll calls to automated badge-based systems, yet institutional environments continue to grapple with fragmented data that fails to deliver strategic insights. In modern universities and professional settings, badge logs suffer from hardware-induced noise, duplicate scans, and proxy entries, limiting their utility beyond basic compliance reporting. The Agent-Based Intelligent Attendance and Workflow Tracking System emerges from this context, leveraging multi-agent AI frameworks like Smol Agents to preprocess noisy data, detect anomalies, and uncover cohort patterns through graph-based analysis.

### 2.2. Importance of Workflow Tracking

Precise workflow tracking through attendance data serves as a leading indicator of participant engagement, performance outcomes, and operational efficiency in both academic and professional institutions. By automating anomaly detection and clustering, institutions can shift from reactive administration to proactive interventions, correlating attendance patterns with retention rates and productivity metrics. This capability significantly reduces the administrative burden, enabling supervisors to prioritize high-value tasks such as mentoring and resource allocation over manual data reconciliation.

### 2.3. Scope of the Project

This project delivers a production-ready prototype encompassing data ingestion from badge logs and iCalendar schedules, agent-orchestrated validation, Louvain-based cohort discovery, and a React dashboard for interactive insights, targeted for deployment in universities and corporate training programs. Excluded are biometric verification or real-time facial recognition due to privacy constraints, with focus instead on explainable AI analytics from existing badge infrastructure [4]. The scope prioritizes scalability via FastAPI endpoints and persistent memory, ensuring seamless integration into institutional workflows without extensive hardware upgrades.

## 3. Problem Statement

### 3.1. Current Challenges

Existing badge-based attendance systems in institutional settings produce raw logs that are inherently noisy, featuring duplicate scans, hardware malfunctions, missed entries, and proxy scanning where participants badge for absent peers. These datasets lack preprocessing to align with scheduled events from iCalendar sources, resulting in incomplete session records and unvalidated anomalies such as out-of-schedule check-ins or device inconsistencies. Traditional analytics fail to model individual behavioral baselines or relational cohort dynamics, treating participants as isolated entities rather than interconnected groups with shared patterns.

### 3.2. Impact on Efficiency and Decision-Making

The absence of interpretable insights forces administrators to conduct manual data reconciliation, diverting substantial time from strategic tasks like early intervention for at-risk participants or resource optimization. This reactive posture delays identification of engagement risks, correlates with diminished performance outcomes, and undermines institutional accountability metrics tied to attendance compliance. Ultimately, unprocessed data erodes the strategic value of attendance tracking, perpetuating inefficiencies in workforce allocation and hindering data-driven governance across professional and academic environments.

## 4. Objectives

### 4.1. Project Goals

The primary goal of this project is to elevate attendance tracking from a passive compliance function to an intelligent decision-support platform deployable across universities and professional institutions. By integrating multi-agent AI orchestration with graph-based analytics, the system transforms noisy badge data into actionable insights that enable early intervention, cohort-level risk identification, and data-driven governance. This initiative positions the prototype as a scalable product that aligns operational data with strategic institutional priorities, fostering measurable improvements in engagement and efficiency.

### 4.2. Functional Objectives

Develop agent-driven analytics capable of learning individual attendance behaviors, detecting anomalies through rule-based validation, and delivering transparent query responses via specialized sub-agents for data preprocessing, cohort clustering, and insight generation. Implement Louvain algorithm-based community detection to uncover hidden attendance patterns and group participants dynamically, complemented by a React dashboard featuring real-time visualizations of sessions, alerts, and engagement matrices. Ensure seamless integration of badge logs with iCalendar schedules through autonomous background processing, producing explainable outputs that support natural-language interactions without exposing underlying computational logic.

### 4.3. Operational Objectives

Achieve production-grade scalability through a FastAPI backend with hot-reload configuration, persistent memory management, and a modular microservices architecture that supports independent component evolution and minimal-downtime deployment. Enable autonomous operation through event-triggered workflows and insight-first resolution policies that prioritize memory reuse over recomputation, reducing latency while preserving context across multi-turn interactions. Deliver measurable operational gains, including a substantial reduction in manual analysis effort and improved advisor efficiency through interactive, data-grounded decision support tailored to institutional workflows.

## 5. Literature and Market Overview

### 5.1. Existing Workflow Management Systems

Existing institutional workflow systems are designed to support administrative coordination, task routing, and attendance-related record keeping across organizations. In educational and professional settings, these systems are often embedded in learning management platforms, student information systems, or workflow automation tools that streamline routine processes such as attendance capture, notifications, approvals, and reporting. While they improve operational efficiency and reduce manual effort, most of these solutions remain focused on data collection and process standardization rather than analytical interpretation. As a result, they provide limited support for detecting abnormal attendance behavior, identifying recurring patterns, or generating actionable insights for decision-makers.

### 5.2. Limitations of Traditional Approaches

Manual and LMS-based systems remain labor-intensive and error-prone, particularly in large-scale or multi-campus deployments, offering only descriptive statistics without predictive capabilities or behavioral modeling. Badge/RFID solutions, while cost-effective for rapid logging, generate noisy data susceptible to proxy attendance, hardware failures, and duplicates, lacking preprocessing against schedules or cohort-level analysis. Dashboard-oriented platforms provide retrospective visualizations via static thresholds but fail to capture evolving patterns, relational dynamics, or data quality issues, forcing manual interpretation that delays interventions.

### 5.3. Emerging AI-Driven Solutions

Emerging AI-driven solutions are increasingly moving beyond static automation toward adaptive systems that combine multi-agent orchestration, persistent memory, and explainable decision support. In attendance and institutional workflow contexts, recent developments include AI-based face-recognition systems for real-time attendance capture, multiple-agent frameworks for coordinating complex operational tasks, and graph-based methods such as the Louvain algorithm for discovering meaningful cohorts or communities [5], [6]. These approaches are particularly relevant because they reduce manual processing, improve interpretability, and support more responsive institutional decision-making. Compared with traditional systems, they enable a shift from simple record keeping to intelligent analysis, where attendance data can be interpreted, grouped, and monitored in a more dynamic and context-aware manner.

## 6. Proposed Solution : Agentic Tracking System

### 6.1. Concept Overview

The Agentic Tracking System constitutes a multi-agent AI platform that orchestrates specialized agents to transform raw, noisy badge-based attendance logs into actionable institutional insights, deployable across universities and professional environments. At its core, a central Orchestrator agent, constructed on the Smol Agents framework, employs an insight-first resolution policy—prioritizing memory reuse before invoking sub-agents for data pipeline processing, validation, Louvain clustering, and analytical insights. This architecture delivers a React-based decision-support dashboard featuring real-time visualizations of anomalies, cohort engagement matrices, and session alerts, ensuring seamless integration with existing badge infrastructure without requiring hardware modifications [3].

### 6.2. Core Principles

- **Automation** : Background schedulers monitor iCalendar events to trigger autonomous data ingestion and agent workflows post-session, eliminating manual reconciliation of badge logs against schedules while applying rule-based validation to filter hardware noise, duplicates, and out-of-sequence entries.
- **Intelligence** : Specialized sub-agents enable adaptive behavioral modeling, anomaly detection, and graph-based cohort discovery via the Louvain algorithm, producing explainable, query-responsive insights that surface individual risk patterns and collective trends without static thresholds or retrospective analysis.
- **Modularity** : Microservices-inspired separation of dashboard, FastAPI endpoints, agent orchestration, and shared memory utilities supports independent scaling and hot-reload configuration, facilitating extensibility for diverse institutional workflows while maintaining thread-safe, context-aware multi-turn reasoning.

## 7. System Architecture

### 7.1. Architecture Overview

The Agentic Tracking System employs a microservices-inspired, multi-layered architecture that separates concerns across dashboard, API, agent orchestration, and shared utilities, enabling independent scaling and deployment while maintaining tight coupling between agent-generated insights and user-facing visualizations. A central Orchestrator agent coordinates specialized sub-agents through a tool-calling strategy with insight-first resolution, leveraging persistent memory and background event scheduling to process badge logs against iCalendar schedules autonomously. This design—illustrated in the high-level architecture diagram—facilitates production-grade operation with hot-reload configuration, thread-safe memory management, and unified repository patterns supporting JSONL, CSV, ICS formats for seamless institutional integration.

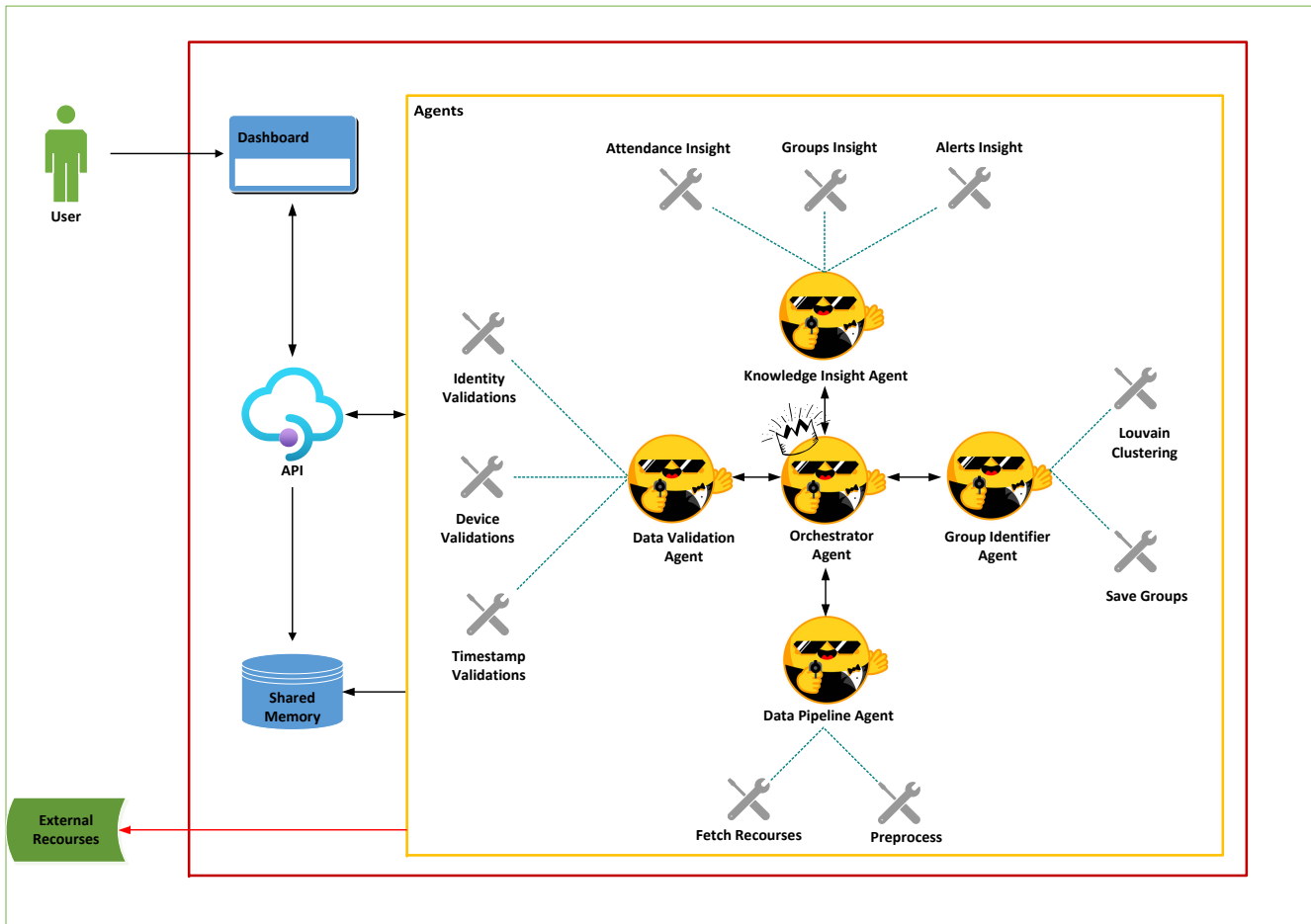


FIGURE 1 – High Level System Architecture Diagram

### 7.2. Main Components

- **Dashboard** : React 18 + Vite + Tailwind frontend renders glassmorphism-inspired D3 visualizations of engagement matrices, session details, anomaly alerts, and cluster memberships, consuming FastAPI endpoints to deliver real-time, interactive decision support tightly synchronized with agentic analytics.
- **Backend (API)** : FastAPI-powered REST endpoints deliver fully documented interfaces for attendance alerts, group analytics, and system health monitoring, exposing agent outputs through structured schemas while abstracting data ingestion, validation, and orchestration workflows for external dashboard consumption.
- **Agent Layer** : A modular multi-agent architecture where each agent is responsible for a specific stage of the analytical workflow.
  - **Orchestrator Agent** : Coordinates the workflow, routes tasks to the appropriate sub-agents, and synthesizes the final response.
  - **Data Pipeline Agent** : Collects, preprocesses, and structures raw attendance data for downstream analysis.
  - **Data Validation Agent** : Checks data integrity and detects inconsistencies such as duplicates, missing records, or invalid timestamps.

- **Group Identification Agent** : Applies Louvain clustering to discover attendance-based cohorts and hidden group patterns.
- **Knowledge Insight Agent** : Generates analytical answers and summaries from attendance, group, and alert data.
- **Memory System** : Chat-scoped short-term memory with dedicated MemoryManager maintains context across multi-turn workflows, enabling insight reuse and reducing recomputation while supporting dynamic configuration via environment variables and config.json for operational tuning without downtime [7].

## 8. Key Features and Functionalities

### 8.1. Intelligent Workflow Automation

The Agentic Tracking System implements multi-agent orchestration through a central Orchestrator that applies insight-first resolution policies, routing tasks to specialized sub-agents for data validation, Louvain clustering, and analytical insight generation while minimizing redundant computation. This enables autonomous processing of badge logs against iCalendar schedules, with rule-based validation automatically filtering hardware noise, duplicate scans, and out-of-sequence entries to produce clean datasets for decision support. Workflow execution remains fully explainable, delivering query-responsive insights that adapt to institutional priorities without manual intervention.

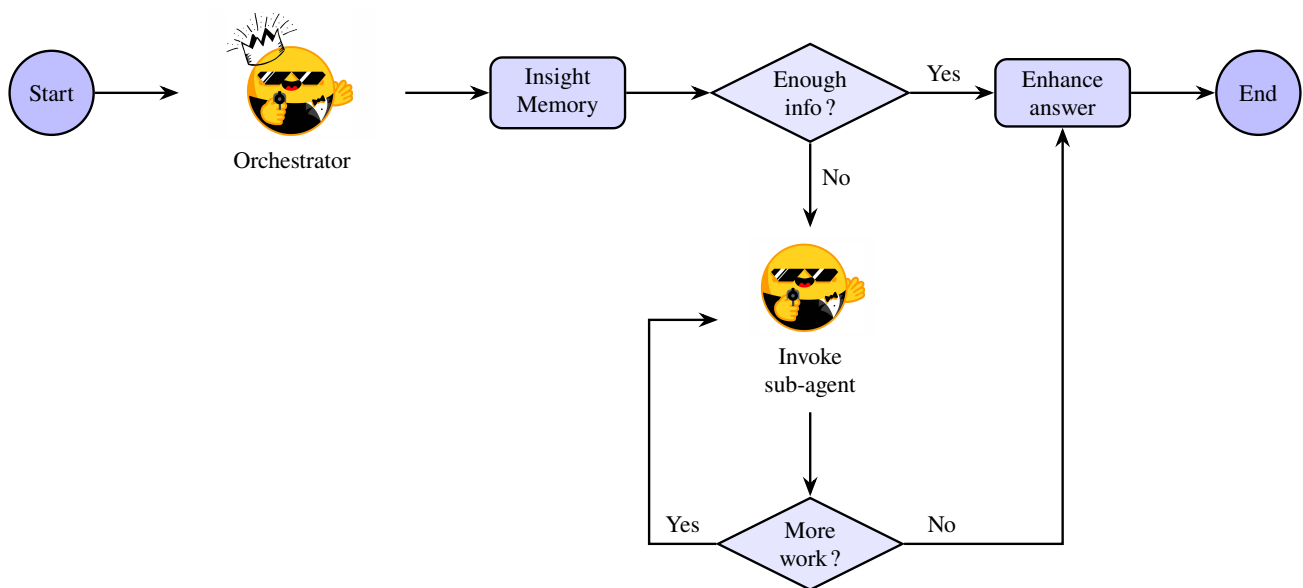


FIGURE 2 – Intelligent Workflow Automation

### 8.2. Context-Aware Memory System

A dedicated MemoryManager maintains chat-scoped short-term memory across multi-turn interactions, preserving analytical context and enabling insight reuse to reduce latency while supporting thread-safe operations in production environments [7]. The system prioritizes existing cached insights before escalating to full agent workflows, ensuring context coherence for cohort analysis, anomaly tracking, and longitudinal attendance pattern recognition. Dynamic configuration via hot-reload parameters allows memory retention policies to adapt to varying institutional data volumes without service interruption.

### 8.3. Real-Time Monitoring Dashboard

The React-based Deep Ocean dashboard delivers glassmorphism-inspired D3 visualizations including engagement matrices, session details, cluster memberships, and triaged anomaly alerts (device, identity, timestamp), providing advisors with interactive, agent-synchronized views of attendance intelligence. Pages for Overview, Groups, Sessions, and Alerts enable drill-down analysis from cohort trends to individual risk profiles, with all metrics dynamically refreshed from FastAPI endpoints to eliminate reporting delays. System health indicators and API documentation ensure operational transparency for institutional administrators.

### 8.4. Background Event Processing

A persistent background scheduler continuously monitors iCalendar events, triggering autonomous Orchestrator workflows precisely when sessions conclude to reconcile badge data against scheduled contexts without human oversight.



This event-driven architecture processes noisy real-world logs through sequential Data Pipeline → Validation → Grouping → Insight pipelines, producing validated analytics within minutes of session completion. Configurable parameters govern processing windows and alert thresholds, supporting diverse institutional schedules and scale requirements.

### 8.5. Flexible Data Handling

Unified Repository Pattern abstracts ingestion across JSONL, JSON, CSV, and ICS formats, enabling seamless integration with existing badge infrastructure and calendar systems common in universities and professional institutions. Rule-based validation agents systematically address prevalent data quality issues—outsiders, device anomalies, redundant UIDs, temporal inconsistencies—transforming fragmented logs into structured sessions ready for graph-based analysis. The modular design supports extension to additional data sources while maintaining backward compatibility with legacy attendance deployments.

## 9. Use Cases and Applications

### 9.1. Natural-Language Investigative Queries

The most distinctive use case of the Agentic Tracking System is the ability to investigate attendance-related behavior through natural-language questions. Rather than requiring users to navigate predefined dashboards or manual filters, the platform allows institutional stakeholders to ask direct questions such as which cohorts show irregular participation, which sessions contain suspicious patterns, or which participants are associated with repeated anomalies. The agents then interpret the request, analyze the available data, and return an immediate answer grounded in the recorded attendance evidence. This interaction model transforms the system from a passive reporting environment into an investigative interface that supports exploratory analysis, rapid clarification, and decision-making in real time.

### 9.2. Administrative Workflow Automation

Institutional administrators can rely on automated validation agents to triage device, identity, and timestamp anomalies extracted from badge data, thereby streamlining audit preparation and compliance reporting across organizational units. The insight-oriented Orchestrator routes complex queries to specialized sub-agents and produces explainable summaries of cohort behavior and session compliance that can be integrated directly into administrative dashboards. By shifting processing to the background, the system reduces manual reconciliation and enables proactive governance based on validated analytics rather than raw event logs. This improves operational consistency while preserving traceability and interpretability.

### 9.3. Cohort Engagement Monitoring

The Agentic Tracking System also supports the examination of participation patterns within project or instructional cohorts by applying Louvain-based clustering to attendance data [6]. This enables supervisors to identify low-engagement groups, irregular participation, and absenteeism trends across multiple sessions. Advisors can consult engagement matrices and anomaly alerts to detect emerging risks early, while also relating attendance behavior to milestone deadlines synchronized through iCalendar integration. In this way, the system provides a structured basis for monitoring participation at the cohort level and for informing targeted follow-up actions.

### 9.4. Event and Deadline Monitoring

The system also supports the supervision of scheduled events and operational deadlines through autonomous background workflows triggered at iCalendar endpoints. Attendance records are continuously compared with planned sessions in order to detect out-of-sequence scans, proxy attempts, and no-show patterns before deadlines expire. Real-time alerts highlight critical anomalies, such as repeated credential use or synchronized absences, allowing program directors to intervene while the event cycle is still active. This capability is especially relevant for institution-wide seminars, mandatory training sessions, and other time-bound activities where attendance integrity is essential.

## 10. Implementation and Methodology

### 10.1. Development Approach

The Agentic Tracking System follows a product-oriented, iterative development methodology emphasizing microservices separation between agent orchestration, API layer, data utilities, and frontend dashboard to enable independent evolution and scaling. Implementation adopts an insight-first resolution strategy where the central Orchestrator agent validates scope before routing tasks through sequential Data Pipeline → Validation → Grouping → Insight workflows, minimizing redundant computation via memory reuse. This agent-centric approach—built on Smol Agents framework—prioritizes explainability, security guardrails, and production-grade reliability through strict task specifications, failure isolation, and hot-reload configuration management.

### 10.2. Tools and Technologies

Backend leverages FastAPI for high-performance REST endpoints with automatic OpenAPI documentation, exposing structured schemas for attendance alerts, cohort analytics, and system health metrics while abstracting complex agent coordination. The agent layer utilizes Smol Agents framework with specialized sub-agents (Orchestrator, Data Pipeline, Validation, Group Identifier, Knowledge Insight) implementing tool-calling patterns and Louvain clustering via NetworkX for graph-based cohort discovery. Frontend employs React 18 + Vite build system + Tailwind CSS for glassmorphism-inspired D3.js visualizations, complemented by unified Repository Pattern supporting JSONL/CSV/ICS ingestion and thread-safe MemoryManager for context persistence.

### 10.3. Deployment Overview

The system supports containerized deployment through Docker Compose for local development and production-grade scaling via Kubernetes orchestration, with background schedulers monitoring iCalendar events to trigger autonomous workflows post-session completion. Hot-reload configuration service—backed by config.json and environment variables—enables parameter tuning (data paths, LLM fallbacks, processing windows) without downtime, while RAGaRenn integration provides sorted LLM model failover for resilience against model deletions [8]. Observability includes comprehensive API documentation, agent execution traces, and dashboard health indicators, facilitating institutional IT integration with existing badge infrastructure and minimal operational overhead.

### 10.4. Institutional Deployment Model

#### 10.4.1. Validated Pilot Experience

The Agentic Tracking System underwent a six-month pilot at IMT Atlantique Rennes Campus using a single portable badge reader device, monitoring attendance across three student cohorts totaling approximately 50 participants. This mobile testing approach—where the device was carried between sessions—successfully validated core agentic workflows, anomaly detection, and dashboard functionality while generating realistic noisy datasets for system refinement.

#### 10.4.2. Production Deployment Strategy

Full institutional deployment requires fixed badge reader installation at entrances to all target teaching spaces—classrooms, laboratories, seminar rooms, and training halls—to enable automated, dependency-free data capture. Each permanently mounted reader serves as a session-specific capture point where participants register presence upon entry, eliminating manual device handling while the backend autonomously aligns logs with iCalendar schedules for validation, clustering, and real-time dashboard visualization.

#### 10.4.3. Scalable Rollout Path

The pilot confirmed technical readiness for distributed deployment across 3-5 priority rooms initially, scaling methodically to institution-wide coverage. This fixed-infrastructure model ensures operational reliability, preserves existing workflows, and delivers comprehensive attendance intelligence applicable to any academic or professional environment requiring precise, hands-free engagement tracking

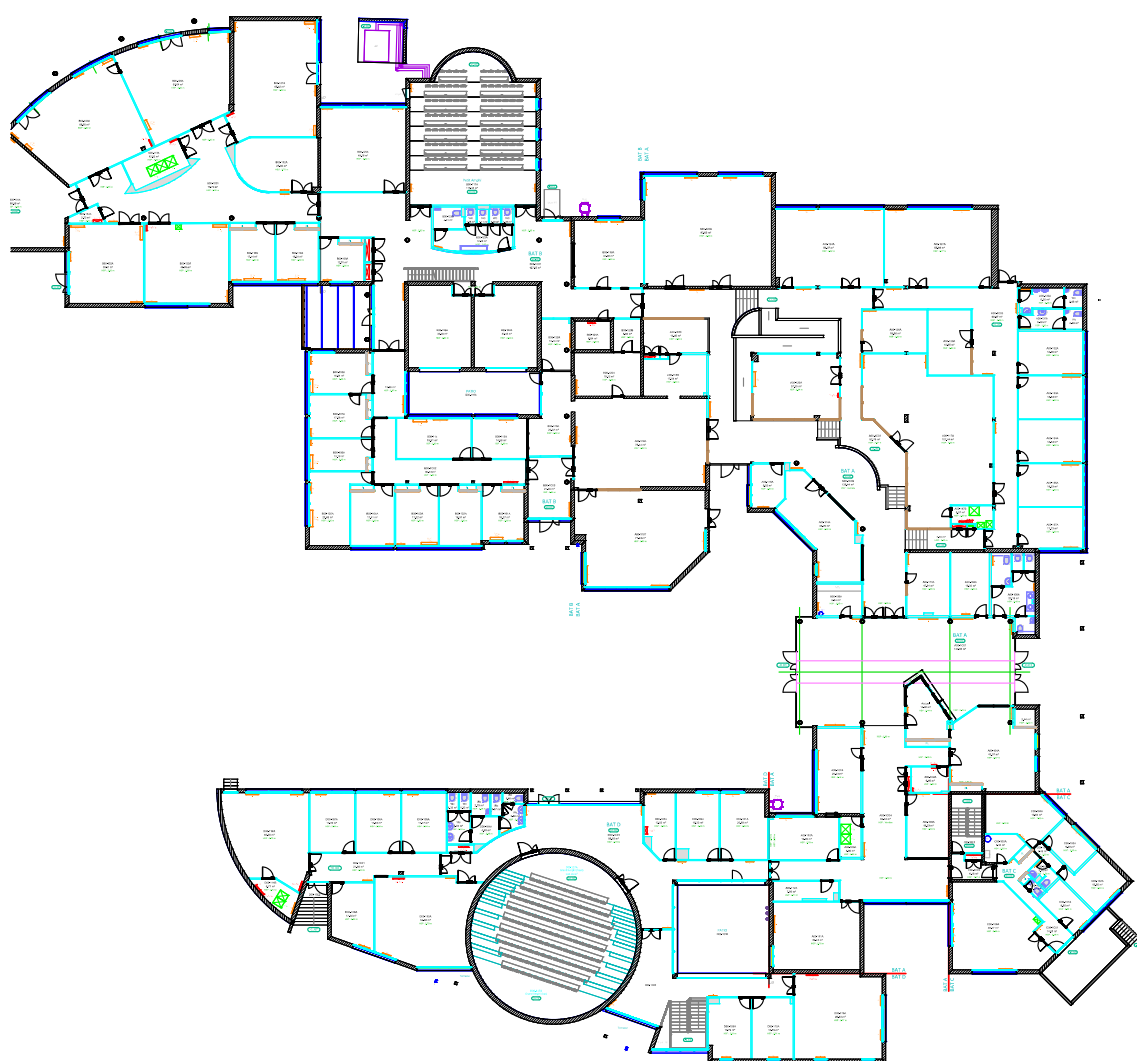


FIGURE 3 – IMT Atlantique Rennes Campus

## 11. Evaluation and Results

### 11.1. System Testing

The system was evaluated over a six-month pilot at IMT Atlantique on a study cohort comprising three student groups and approximately 50 students. Comprehensive testing validated the multi-agent orchestration across the full workflow, from data ingestion and validation to Louvain-based clustering and knowledge-insight generation. The test campaign confirmed that the pipeline executed successfully on synthetic attendance data containing realistic noise, including duplicate records and out-of-sequence scans. Unit-level verification also confirmed that the specialized agents operated independently as intended, particularly in relation to device, timestamp, and identity anomaly handling. End-to-end integration tests further showed that the dashboard remained synchronized with the API layer, with front-end views accurately reflecting the outputs generated by the orchestration layer.

### 11.2. Performance Evaluation

The insight-first orchestration strategy improved responsiveness by reusing stored context whenever relevant information was already available, thereby reducing recomputation. This behavior supported low-latency interaction for repeated analytical queries and made the system suitable for cohort-level exploration in a production-like setting. The Louvain clustering component remained appropriate for scalable community detection in attendance graphs, consistent with the established properties of the method in large networks [6]. The backend architecture was also designed to support concurrent request handling and conversational context across extended multi-turn interactions.

### 11.3. Observed Outcomes

In pilot use, the system surfaced actionable attendance patterns, including low-engagement clusters, and recurring anomalies across sessions. The categorized alerts reduced the effort required to review attendance irregularities and supported faster triage of device, identity, and timestamp issues. The dashboard enabled stakeholders to move from institution-wide summaries to individual-level inspection, improving the practical usefulness of attendance data for operational decision-making. Overall, the results suggest that the system can support proactive intervention and more informed attendance management within an institutional context.

## 12. Value Proposition and Impact

### 12.1. Efficiency Improvement

The Agentic Tracking System improves operational efficiency by prioritizing cached insights before triggering full multi-agent workflows, thereby reducing unnecessary recomputation and supporting faster query resolution. In prototype testing, this insight-first approach enabled rapid cohort analytics and timely processing of attendance-related events after session completion. The API layer was designed to remain responsive under concurrent usage, supporting interactive access for staff without visible degradation in the test environment.

### 12.2. Reduction of Manual Work

The system reduces manual workload by automating validation, anomaly triage, and cohort discovery. Instead of requiring staff to reconcile raw logs manually, specialized agents classify device, identity, and timestamp irregularities into interpretable categories and expose them through a unified interface. The repository structure also minimizes integration overhead by handling heterogeneous input formats through a single ingestion layer, which is especially useful in institutions that manage multiple attendance sources.

### 12.3. Enhanced Decision-Making

The dashboard improves decision-making by translating attendance data into actionable visual summaries, cohort-level patterns, and individual risk profiles. Rather than presenting static attendance percentages, the system highlights recurring anomalies, proxy-like behavior, and low-engagement clusters that can support earlier intervention. This design strengthens the connection between attendance monitoring and institutional retention objectives by enabling staff to focus on patterns that matter operationally.

### 12.4. Scalability and Future Readiness

The modular architecture supports future scaling from pilot deployment to broader institutional use. Its separation of dashboard, API, agents, and shared utilities makes it easier to extend the system with additional attendance sources, including biometric or LMS integrations, while preserving backward compatibility with the existing badge-based workflow [2], [4]. The use of configurable services and modular agent specifications also improves resilience and adaptation as operational requirements evolve.

## 13. Limitations and Risks

### 13.1. Technical Limitations

The current prototype relies on regional LLM services with fallback mechanisms that may introduce processing delays during model unavailability. Louvain clustering performance is optimized for the tested cohort sizes but would require distributed computing infrastructure to scale beyond current pilot deployments. Short-term memory constraints limit analysis to recent sessions unless extended persistence is implemented, while rule-based validation effectively handles common anomalies but requires customization for institution-specific patterns such as shared credentials or irregular schedules.

### 13.2. Integration Challenges

Different badge systems use varying data formats, requiring custom adjustments to import attendance logs smoothly. Calendar synchronization can be disrupted by institution-specific scheduling tools or multi-department calendars. Older badge hardware without modern APIs forces manual file exports instead of live data feeds. Privacy laws also require careful data storage limits that can interrupt long-term analysis workflows.

### 13.3. Adoption Barriers

Academic stakeholders may initially perceive AI-driven anomaly detection as punitive rather than supportive, requiring clear communication that the system identifies operational patterns rather than individual infractions. Faculty unfamiliarity with interactive dashboards represents a training challenge, as advanced visualizations require some technical orientation despite their intuitive design. Privacy concerns persist around anonymized badge data, particularly regarding potential correlation with external identifiers, necessitating transparent data handling policies and governance frameworks.

## 14. Future Work

### 14.1. Role-Based User Management

Implement hierarchical access controls enabling department heads to query only their departmental attendance data while institutional management accesses organization-wide analytics. Role-specific dashboards and query permissions will enforce organizational policies, ensuring data security and operational alignment across different administrative levels.

### 14.2. Proactive Notification System

Develop automated alert workflows that notify both individuals and their supervisors when attendance patterns indicate risk. The system will support justification workflows for lateness or absence, maintaining audit trails while enabling constructive interventions rather than punitive measures.

### 14.3. Customizable Periodic Reporting

Enable supervisors to define personalized reporting tasks through natural language prompts, with the system generating and delivering scheduled reports automatically. This "Tasks" functionality will support recurring analytics such as weekly cohort trends, monthly anomaly summaries, or custom performance metrics tailored to individual management priorities.

## 15. Conclusion

In summary, the Agentic Tracking System establishes a new standard for institutional attendance management by leveraging multi-agent AI to convert noisy badge logs into precise, actionable intelligence that drives proactive governance and operational excellence. Pilot validation across real-world data demonstrated its capacity to autonomously detect anomalies, discover behavioral cohorts through Louvain clustering, and deliver interactive insights via a scalable React dashboard—capabilities that address longstanding gaps in traditional systems.

This prototype not only reduces administrative workload through automated validation and background processing but also transforms attendance from a compliance metric into a strategic early-warning signal for engagement risks and performance trends, enabling institutions to intervene effectively before issues escalate.

With planned enhancements in role-based access, proactive notifications, and customizable analytics, the system offers a clear path to enterprise deployment, delivering sustained value in student success, resource optimization, and data-driven decision-making across academic and professional environments.

## Références

- [1] S. Nimje, R. Taneja, O. Baviskar, and R. Y. Patil, “ElizaEdu : AI-powered Web3 system for automated, secure event attendance verification in educational institutions,” *Journal of Trends in Computer Science and Smart Technology*, vol. 7, no. 1, pp. 67–83, Mar. 2025, Available : <https://doi.org/10.36548/jtcsst.2025.1.005>.
- [2] S. Pathan, L. Bommireddipalli, B. Deka, S. Anandham, A. T. Bhattu, and S. Boorla, “A location and QR code-based solution for efficient attendance management,” in *Proc. Int. Conf. Mach. Learn. Auton. Syst. (ICMLAS)*, 2025, pp. 859–866, [Online]. Available : IEEE Xplore, Part Number : CFP258A9-ART, ISBN : 979-8-3315-0574-5.
- [3] M. F. Makkawi, “Agentic\_tracking\_system : AI-driven attendance and workflow tracking system that converts noisy badge data into actionable insights for institutions, improving operational visibility, anomaly detection, and decision-making,” GitHub repository, [Online]. Available : [https://github.com/MoMakkawi/Agentic\\_tracking\\_system](https://github.com/MoMakkawi/Agentic_tracking_system) Accessed : May 16, 2026.
- [4] I. Kakepoto, N. A. Bhutto, and A. I. Umrani, “Biometric attendance management system : A university perspective,” *Global Social Sciences Review*, vol. VII, no. II, pp. 309–318, 2022, doi : 10.31703/gssr.2022(VII-II).31.
- [5] S. Hemavathi and R. Chakravarthi, “AI-powered security and attendance management system using deep learning and facial recognition,” *Journal of Information Systems Engineering and Management*, vol. 10, no. 35, 2025. [Online]. Available : <https://www.jisem-journal.com/>
- [6] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, “Fast unfolding of communities in large networks,” *Journal of Statistical Mechanics : Theory and Experiment*, vol. 2008, no. 10, p. P10008, Oct. 2008. [Online]. Available : <https://doi.org/10.1088/1742-5468/2008/10/P10008>
- [7] J. Liang *et al.*, “AI meets brain : A unified survey on memory systems from cognitive neuroscience to autonomous agents,” *ACM Transactions*, vol. 1, no. 1, pp. 1–57, Dec. 2025, doi : 10.1145/nnnnnnn.nnnnnnn. [Online]. Available : <https://github.com/AgentMemory/Huaman-Agent-Memory>
- [8] RAGaRenn : Plateforme d’IA générative open source, Eskemm Numérique (GIP Numérique de Bretagne). [Online]. Available : <https://ragarenn.eskemm-numerique.fr/> Accessed : Jan. 2026.

